MMReD_A Cross-Modal Benchmark For Dense Context Reasoning

Abstract

MMReD_A Cross-Modal Benchmark For Dense Context Reasoning, p.1

> ability to perform complex multi-modal reasoning over extended contexts remains critically limited.

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assess reasoning abilities within dense, information-rich scenarios where simple retrieval is not enough

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> MMReD challenges models to identify and interpret global patterns across entire context

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> Our benchmark comprises 24 tasks of varying complexity, ranging from standard passkey retrieval setups to those requiring selective or uniform attention to all context chunks.

Benchmarks are aiming for diversity.

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> The evaluation reveals a consistent performance drop across all tested models – including the most advanced LLMs, LVLMs, and architectures specializing in code and reasoning – as the number of observations increases.

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> Conventional fine-tuning techniques, such as SFT and GRPO, also fail to generalize effectively to longer contexts. These observations reveal an inherent limitation in current model architectures, emphasizing the need for innovative approaches to enable competent dense context reasoning in multi-modal AI systems.

Authors claim RL techniques provide no added benefit for long-context reasoning.

Conclusion

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> Our results demonstrate that current LLMs and LVLMs struggle significantly with long-context understanding, with performance degrading rapidly as context length increases beyond 32 images.

Do we also observe this phenomenon? What about interleaved image and text?

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> Model size positively correlates with long-context performance, but the benefits diminish at larger scales, suggesting architectural limitations rather than parameter count as a key bottleneck. Additionally, our analysis of multimodal adapter types and frame pooling strategies revealed that the architectural components choice critically affects model's ability to reason over long visual sequences.

Authors also generally do an architecture analysis

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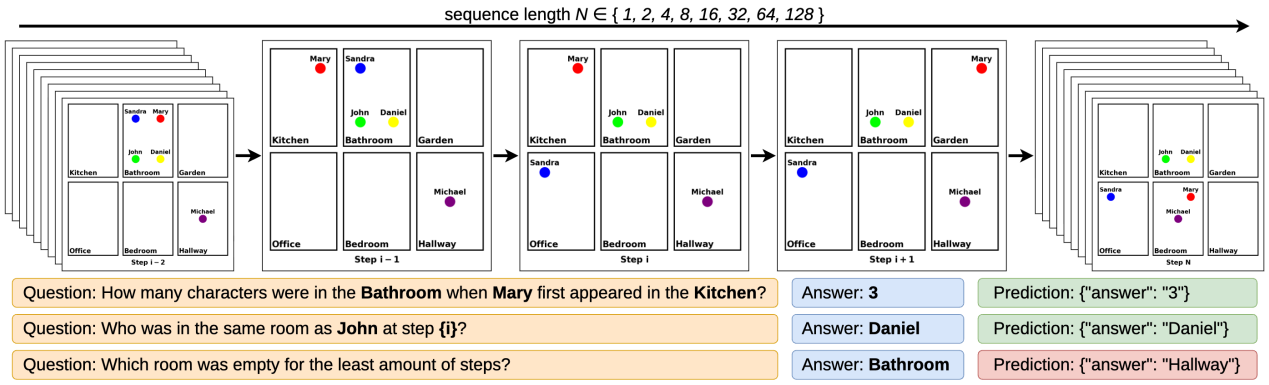
> Unexpectedly, video-oriented LVLMs and multimodal instruction tuning are not beneficial for longcontext reasoning. The observed performance decline for these models on extended sequences suggests that current video pretraining does not sufficiently address the challenges of long-context tasks.

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> Our ablation studies confirmed that MMReD's conclusions are robust to potential confounding factors such as the lost-in-the-middle phenomenon. By maintaining unique and randomly evolving environments, the benchmark ensures that performance degradation reflects genuine limitations in long-context reasoning rather than artifacts of data structure or task formulation.

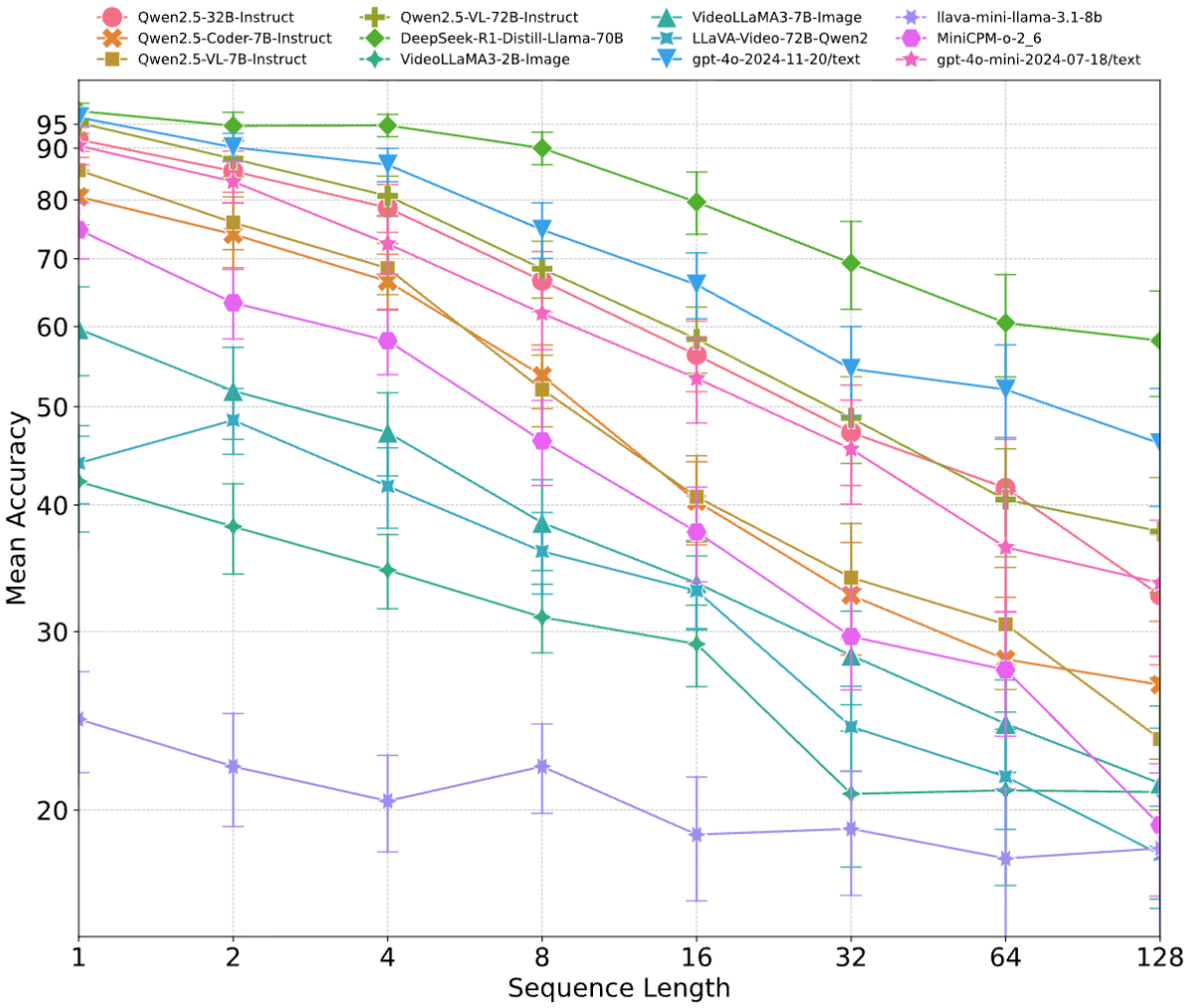
Authors confirm the popular phenomenon like lost-in-the-middle even for latest VLMs through rigorous testing.

Tables and Figures

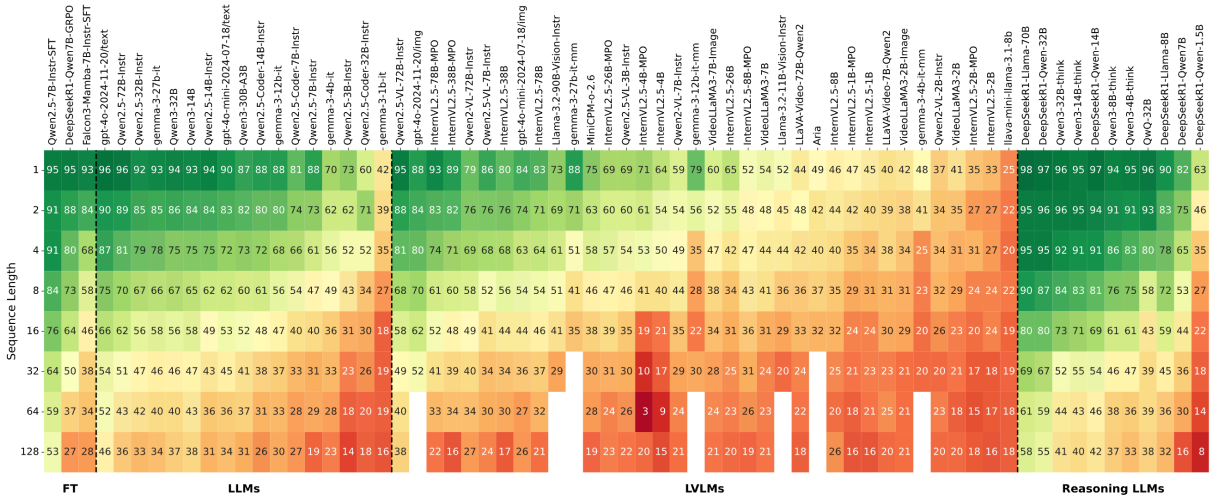


Premise

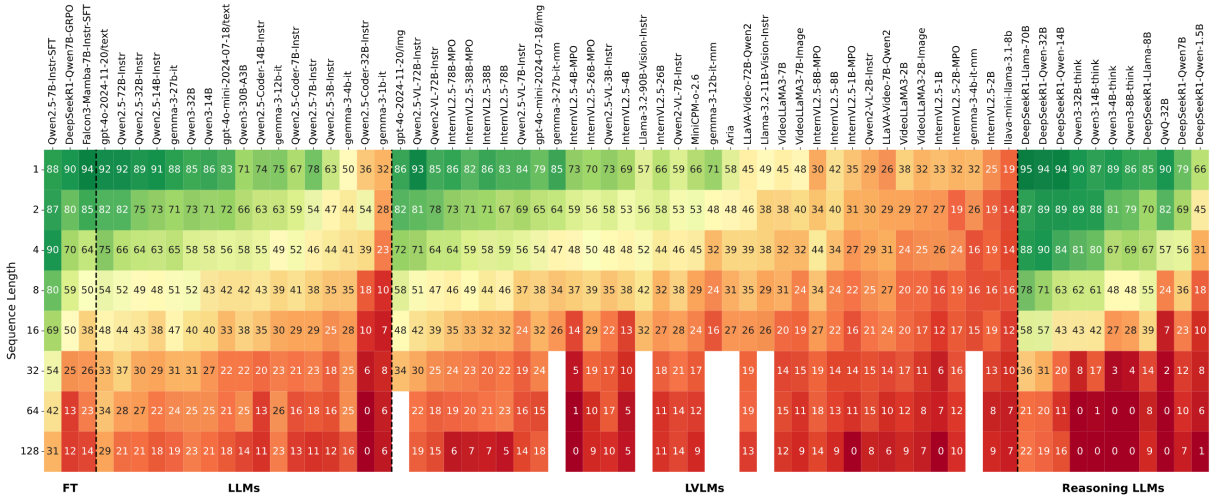
- The model is given an input of [1, 128] images.
- The model is asked questions based on this sequence.
- The sequence of images appears to related and sequential, with the model expected to keep track of entities in this sequence of images.



- Authors test models upto 72B
- This task appears more complex than ours. If we are able to achieve accuracy drop for simpler tasks, then its a good finding.
- NOTE: For big models, scores do not drop below 60 at 128 context length



(a) Models' mean accuracy calculated for all MMReD tasks (24 questions).



(b) Models' mean accuracy calculated only for dense context (DC) type of tasks.

Figure 3: Models' mean accuracy on MMReD across all sequence lengths, grouped by model type: fine-tuned LLMs, general-purpose LLMs, LVLMs, and reasoning LLMs. Cell color indicates performance (green – high, red – low), with models ranked by average score within each group.

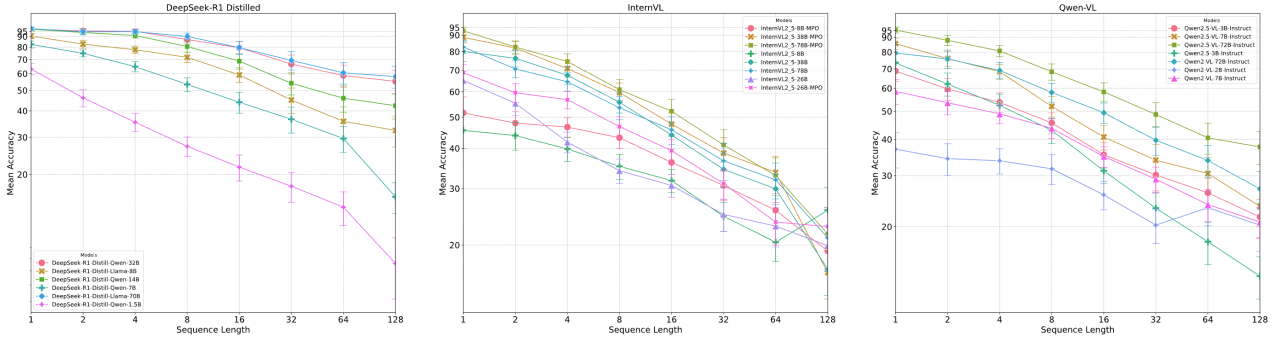
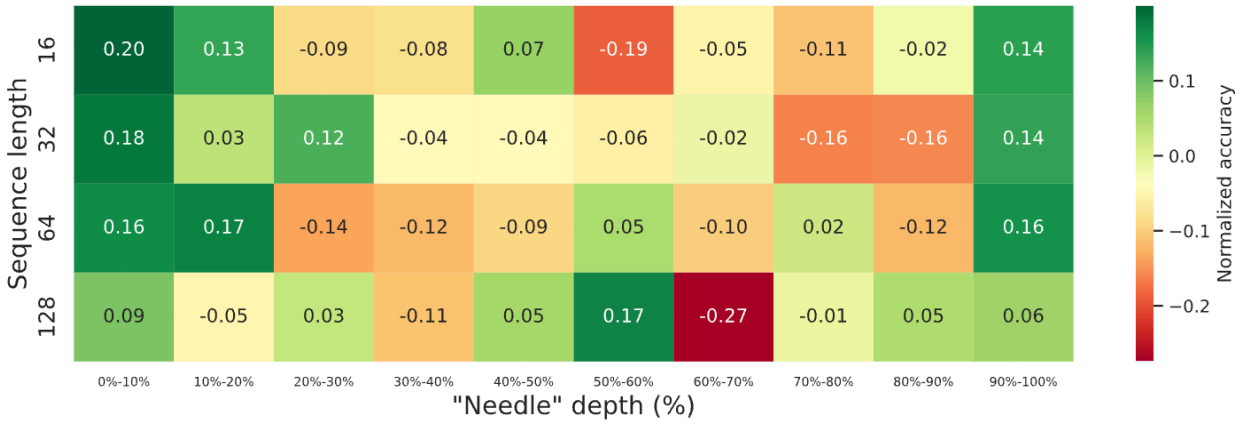


Figure 5: Impact of the number of model parameters on the ability to successfully infer a response using a long visual or text context from MMReD.

Authors have done a nice model family-wise accuracy vs model parameter comparison.

- Higher params have got higher scores.



- Authors observe "lost-in-the-middle" phenomenon.

Introduction

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> Standard practice on reasoning evaluation is to rely on a range of public benchmarks that test reasoning capabilities on a variety of topics. Short-context reasoning includes general language understanding and reading comprehension (DROP (Dua et al., 2019), ARC (Clark et al., 2018), BBH (Suzgun et al., 2022)), academic reasoning (MMLUPRO (Wang et al., 2024d), GPQA (Rein et al., 2024), MATH (Hendrycks et al., 2021), AIME (MAA, 2024)) and code comprehension (HumanEval (Chen et al., 2021), CRUX (Gu et al., 2024), MBPP (Austin et al., 2021)). Long-context reasoning includes recent benchmarks RULER (Hsieh et al., 2024), BABILong (Kuratov et al., 2024), and Michelangelo (Vodrahalli et al., 2024).

For further review - context based and reasoning based benchmarks

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> Large vision-language models (LVLMs) are also rapidly closing the gap in performance on various visual recognition tasks that require reasoning in both text and visual modalities. Existing benchmarks can be categorized by modality of context in the question: single image understanding (MMMU (Yue et al., 2024a;b)), multiple image understanding (MuirBench (Wang et al., 2025), BLINK (Fu et al., 2024b)), and video understanding (VideoMME (Fu et al., 2024a), MLVU (Zhou et al., 2024), EgoSchema (Mangalam et al., 2023), LVBench (Wang et al., 2024b)).

Multimodal benchmarks

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> While these benchmarks demonstrate improved performance on longer contexts, they primarily focus, as we show below, on retrieval-based or Needle-in-a-Haystack (NIAH) setups, where models locate a specific fact from an otherwise irrelevant or distractor-filled context. However, such tasks do not fully capture a model's ability to reason across densely distributed information. In our analysis, we show that state-of-the-art LLMs and LVLMs exhibit no clear correlation between their NIAH performance and their ability to perform deeper, structured reasoning in the information-rich scenarios. So NIAH performance on existing benchmarks alone is not a reliable indicator of reasoning ability.

Authors claim previous NIAH benchmarks have no reasoning involved.

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> To bridge this gap, we introduce MMReD (Multi-Modal REasoning in Dense context), a benchmark designed to assess a conceptually different capability: reasoning in dense environments where models must attend uniformly across the entire context.

Authors propose a benchmark that makes the model contribute equal attention to the entirity of the context.

Method

Design principle

- Minimalist setup with less visual clutter.
- Minimal linguistic construction - focus on reasoning rather than prompt/instruction understanding. We do the same.

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> Our benchmark is designed to scale in complexity on the context length axis. The dataset includes sequences of varying lengths, with longest sequences reaching 128 frames, where even the current best models start to

> systematically score 0% accuracy in some tasks. With the increasing capabilities of models, the benchmark can be extended to larger contexts (e.g., 256 frames or beyond), ensuring that it remains a dynamic and evolving evaluation framework.

We could go upto 128 distractors

Dataset

- Environment
  - 6 distinct rooms and 5 characters
- Input is a sequence of this environment over a period of time.
  - Characters change rooms throughout the sequence.
- Each image in the input sequence has a state number associated with it

Tasks

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> MMReD consists of two primary categories of questions: scene-referenced (NIAH) and dense context (DC) reasoning questions. Each category is designed to systematically evaluate different aspects of multi-modal reasoning

2 tasks proposed by authors

MMReD_A Cross-Modal Benchmark For Dense Context Reasoning, p.5

> NIAH questions. The first three sections of Table 1 correspond to the NIAH task. These questions are designed to be answered based on a single image within the sequence that satisfies the condition

Correspond to single-hop questions in our setting.

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> We created three subgroups of questions: First Appearance (FA), locating the earliest occurrence of a given entity or event; Final App (FI), focusing on the last occurrence; Frame X (FX), focusing on an explicitly specified frame. This division allows us to isolate and measure the lost-inthe-middle phenomenon (Wu et al.,

> 2024), where models struggle to retrieve information from intermediate context positions

Lost in the middle setting based on three question types.

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> DC questions. The final section of Table 1 introduces our novel dense context (DC) reasoning tasks. Unlike the NIAH questions, which require retrieval from a single image, these questions demand global and uniform attention to the entire sequence. We similarly diversify DC tasks to include the same reasoning categories and output types, resulting in nine question types.

Multi-image reasoning questions. Similar to multi-hop.

Results

Main results

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> The performance of all evaluated models begins to degrade significantly for sequence lengths exceeding 32 steps

Performance degrades when 32 images are provided as inputs.

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> Larger models demonstrate a greater ability to maintain accuracy in constructing logical chains over longer sequences

Effect of params on accuracy - bigger models higher accuracy (not exciting, expected)

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> Furthermore, reasoning-specialized LLMs not only exhibit superior initial performance but also show greater robustness to increases in the length of the reasoning chai

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> Multimodal instruction tuning impairs long-context understanding. Additionally, LVLMs struggle to utilize visual context effectively, even when the context length remains within the claimed supported limits. For instance, InternVL2.5, which reports a context length of 16,384 tokens and uses 256 tokens to encode a single image, should handle tasks involving up to 64 images with optimal accuracy. But we observe a decline in performance starting from 16 images.

YESS, same observation

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> mpact of LLM reasoning on long-context retention. Proprietary GPT-4o and its ‘mini’ version outperform most open-source models in both textual and multi-modal representations of the benchmark

Prop. models outperform open-source models.

Task ablations

MMReD_A Cross-Modal Benchmark For Dense Context Reasoning, p.8

> Isolating DC reasoning. Figure 4a illustrates the breakdown in correlation between performance on retrieval-based (NIAH) and DC reasoning tasks as context length increases. We calculate Pearson correlation of model scores averaged for every task within selected task groups. In the first experiment, we select DC and NIAH groups; in the second experiment, we select FA plus FI and FX groups. While Pearson correlation between different NIAH subsets remains consistently high (around 0.9), the correlation between NIAH and DC performances notably declines to approximately 0.5–0.7 at 32 frames and beyond. Our findings closely align with those reported in BABILong (Kuratov et al., 2024): correlation between the models’ ranking on their long-context benchmark and short-context one drops from 0.9 to 0.6 when context length increases. These findings suggest that models successful in NIAH-style retrieval do not necessarily generalize to the proposed DC reasoning scenarios. And thus MMReD allows assessing a conceptually different capability, which we call dense context reasoning.

Authors provide reasoning for dense-context reasoning - a drop in pearson correlation beyond 32 frames between NIAH and DC task indicate 32 distractors and beyond is

dense context reasoning.

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> Perceptual complexity. Perceptual ambiguity, e.g., occlusion, visual noise, recognition errors, plays a major role in real-world multimodal reasoning. However, MMReD is specifically designed to isolate reasoning capabilities by minimizing perceptual complexity

Toy like setup, real-world is much harder.

Modality ablations

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> We computed the relative performance gap between text-only and multimodal variants as LLM−LVLM LVLM × 100%. The gap varies with model scale and context length: text representations generally outperform multimodal ones at mid context lengths, while smaller models sometimes invert at extreme N

Architecture

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> Both fine-tuned Transformer and Mamba architectures demonstrate the same decline in performance as benchmarked zero-shot models

Decline is architecture agnostic.

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> Additionally, we performed a controlled In-Context Learning (ICL) ablation by appending five solved examples (from N ≤ 16) to each prompt.

Good experiment idea for our paper.

Model parameter count

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> The results from the Qwen2.5-VL and InternVL2.5 series reveal a clear trend: accuracy in question answering improves as the number of parameters increases.

Adapter type

MMReD_A Cross-Modal Benchmark For Dense Context Reasoning, p.9

> Multimodal Adapter Type. We evaluated the efficiency of LVLMs using different adapter types: multi-linear layer adapter (Qwen2.5-VL-7B-Instruct (Bai et al., 2025a)), QFormer (MiniCPM-o- 2.6 (Yao et al., 2024)), and cross-attention (Llama-3.2-11B-Vision-Instruct (Chu et al., 2024)). The results in Figure 11 show that LLaMA does not provide high performance. The results of Qwen2.5VL and MiniCPM are roughly the same: both models perform poorly on DC-CC-I and DC-SR-I questions, but MiniCPM performs better on FI-FA-R ones. Moreover, Qwen2.5 has low scores for short sequences in certain types of questions

Overall, adapter type does not seem to play a major role - indicating need for improvement in all types.

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> Frame Pooling Methods in Video-Specific Models. We also compare performance of the different video-specific models: InternVL-2.5 (Wang et al., 2024c), Aria (Li et al., 2024), VideoLLaMA- 3 (Zhang et al., 2025a), and LLaVA-Video-7B-Qwen2 (Zhang et al., 2024b). As shown in Figure 12, VideoLLaMA-3 provides the best performance. Aria performs better than InternVL-2.5. Moreover, InternVL-2.5 has a considerable drawdown for all sequence lengths in certain types of questions

Could we also extend to video - although does not fit in context.

Future directions

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> Possible directions of solving the challenge of dense context reasoning include, but are not limited to, examining the potential of architectural innovations specifically designed for long-context processing, overcoming problems of the

> attention sinks and similar known expressivity issues of RNNs, and incorporating uncertainty quantification (Fadeeva et al., 2024; 2023; Vashurin et al., 2025) to make reasoning more steerable and not materialize false information in the decoded logits; building on existing approaches of test-time compute scaling alongside rigorous RL goal-based pre-training.

Future directions